

Eslam A. Hugair

Engineer for Data Pipelines, ML Tooling & Infrastructure

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Bio

Engineer with 10+ years building data-ingestion pipelines, internal tooling and the infrastructure around ML systems. I write production Python and TypeScript services that move data through queues (BullMQ, Kafka), persist it across MongoDB / Elasticsearch / vector stores, and surface it to ML and research teams through dashboards, debugging tools and APIs. I own the DevOps side too — Docker, CI/CD, AWS / GCP — so the pipelines I build actually run in production. Comfortable building the platform front-ends that wrap this work, including custom annotation tooling.

Work Experience

Mannar.sa – KSA (Remote)

09/2025 - Present · Part-Time

Sr. Front-end Engineer & DevOps

- About: Saudi Arabia's first online legal-consultation platform; secure, scalable architecture for digital legal services.
- **Achievements / responsibilities:**
 - Maintained UI/UX and the frontend architecture for a high-trust legal platform.
 - Owned platform security, compliance and performance work.
 - Managed infrastructure, monitoring and DevOps operations.
 - Built and tuned CI/CD pipelines to shorten the deploy cycle.
 - Scaled the platform from pilot to thousands of active users.
 - Designed and shipped the notification system.
 - Implemented SEO scripts and post-build meta-tag injection.
- Highlighted technologies used: **TypeScript, Node.js, React, AWS, Supabase, Firebase, MongoDB, CI/CD.**

Revic.ai – USA (Remote)

12/2022 - 09/2025

Sr. Full-Stack Engineer / Team Lead

- About: data-driven AI analytics platform surfacing department-level insights from heterogeneous integrations.
- **Achievements / responsibilities:**
 - Built the data-ingestion services that stream from a variety of integration tools into the analytics platform — the team's auditable intake layer.
 - Designed and shipped debugging tools that helped the ML and data teams identify and resolve data-integrity issues.
 - Partnered with the ML team to surface high-integrity model outputs and feedback loops into the product.
 - Authored Azure Functions services for ingestion, enrichment and downstream notifications at scale.
 - Worked with Cognito Search and Elasticsearch for big-data retrieval workloads.
 - Built the Slack bot interface that lets clients explore use cases and act on insights.
 - Promoted to team lead within 6 months for adaptability and ownership in a fast-growing team.
- Highlighted technologies used: **TypeScript, Python, Node.js, React, Azure Functions, CosmosDB, MongoDB, Elasticsearch, Cognito Search, Slack APIs.**

Sr. Front-End Developer

- About: development agency focused on shipping quality products.
- **Achievements / responsibilities:**
 - Recovered a real-estate project with a refactoring plan that reduced technical debt and shortened time-to-feature.
 - Built features and responsive UI components alongside the platform refactor.
 - Improved product performance and developer-experience around the codebase.
- Highlighted technologies used: **TypeScript, Node.js, Next.js, AWS Fargate, Jenkins, Docker, MySQL, GraphQL.**

Full-Stack Engineer → Team Lead → CTO

- About: cart-solution startup focused on boosting conversions with targeted funnels.
- **Achievements / responsibilities:**
 - Built a custom design-system in React to back the product's design and functional requirements.
 - Built utility packages of reusable functionality shared across product services.
 - Owned DevOps for product services (GitHub Actions, AWS) — held 99.9% uptime across the product lifespan.
 - Promoted to Team Lead within a year and CTO the year after.
 - Built a standalone affiliate system to track promoter commissions — idea to production in one month.
 - Built an activity-tracking service for customer events, purchases and lead capture.
 - Designed a payment-integration facade fronting multiple third-party gateways.
 - Helped hire and onboard the team to 5+ members; ran assessments and interviews.
- Highlighted technologies used: **Node.js, React, AWS, GitHub Actions, Netlify, Next.js, TypeScript, MongoDB, PostgreSQL.**

Full-Stack Engineer / Team Lead

- About: platform that augments the Google Ads customer experience with templates and campaign tooling.
- **Achievements / responsibilities:**
 - Designed and built the mobile-friendly web app for Zaino clients to manage campaigns and ship banners.
 - Designed and built the API plus the Google Ads integration services that power campaign report queries.
 - Built the admin console for ad-fee control, audience segments and notification management.
 - Built a Socket.io customer-support dashboard for live communication with customers.
- Highlighted technologies used: **Node.js, React, AWS, PostgreSQL, Google Ads API, Socket.io.**

Selected Projects

A selection of pipelines and tooling I have built solo or led.

VEngine

Solo builder · Data pipelines + AI · 2026

Video knowledge engine — a 4-phase pipeline (ingestion, processing, dendrogram, search index) that transcribes, summarizes, embeds and clusters video into a searchable, clip-level knowledge graph.

- BullMQ pipeline: FFmpeg audio extraction → 5-min chunking → parallel Deepgram / Google STT → assembly → Gemini / GPT-4o summarization → topic + section extraction → 30s sliding-window embeddings → knowledge-graph build → UMAP + agglomerative dendrogram for hierarchical retrieval.

- Swappable providers across STT (Deepgram / Google), LLM (Gemini / OpenAI) and embeddings (Gemini 768d / OpenAI 1536d); A/B comparison built in.
- Per-video and per-user knowledge graphs with cross-video deduplication and SSE progress streaming to the dashboard.
- Containerized, deployed to Google Cloud Run; storage backed by local disk or GCS.

Stack: **NestJS, BullMQ, MongoDB, Redis, Deepgram, Google STT, Gemini, OpenAI, UMAP, GCS, Cloud Run, Turborepo.**

GoValidate

Solo builder · Full-stack + LLM orchestration · 2025 – Present

AI-native startup-validation platform that turns a rough idea into an evidence-backed verdict on demand, competition, risk, and the next move.

- Designed and shipped the platform end-to-end: NestJS API, Next.js 15 console, Caddy + Docker Compose infra.
- BullMQ multi-agent pipeline that mines Reddit signals, generates and scores ideas with multi-provider LLMs (OpenAI + Anthropic via LangChain).
- Hybrid vector + keyword search over Qdrant embeddings; source-credibility scoring (0–3) filters academic / government / press sources.
- Per-job artifact stores, retry semantics and observability across stages of the pipeline.

Stack: **Next.js 15, NestJS, TypeScript, BullMQ, MongoDB, Qdrant, OpenAI, Anthropic, LangChain, Docker, Caddy.**

<https://govalidate.dev>

Denture Mesh + Content Engine

Contributor · Mesh processing + LLM · 2025

Toolkit for dental-mesh capture, annotation and AI-generated educational content for practitioners and patients.

- Mesh annotation pipeline over depth-camera captures using Open3D and NumPy.
- LLM-backed content generation layer for practitioner-facing material; owned the AI side while collaborating on the mesh tooling.

Stack: **Python, Open3D, NumPy, OpenAI, LangChain.**

QuiQ

Lead full-stack engineer · DevOps · Jan 2025 – Present

Internal review, auditing and quality-control platform for SAT content, with feedback, activity tracking, role management and migration tooling.

- Migrated the backend from Netlify Functions to a long-lived Express API to get predictable load distribution and lower cold-start latency.
- Built feedback collection, activity / request tracking, role-based access control (super-admin / admin / user) and an invitation flow.
- Owned the migration tooling that piped curated QuiQ content into the Puef learning platform.
- Handled DevOps end-to-end: environments, CI, monitoring, and the staging / production split.

Stack: **React, TypeScript, Node.js, Express, MongoDB, Auth0, Zod, Netlify.**

<https://quiq.netlify.app/>

Puef.ai

Contributor · LLM endpoints + frontend · Jan 2025 – Present

AI-driven SAT learning platform that adapts planning, review scheduling and content to each learner.

- Hardened the LLM generation endpoints and the 'fixer' flows that detect and repair invalid or unclear model outputs.
- Contributed across the multi-provider LangChain layer (OpenAI / Anthropic / Gemini / Deepseek).
- Worked on the React frontend and the integration glue with the QuiQ review pipeline.

Stack: **Python, Flask, LangChain, OpenAI, Anthropic, Gemini, Deepseek, MongoDB, React, TypeScript, Railway.**

<https://puef.ai>

Story-Teller

Solo builder · LLM orchestration · 2025

Turns a written story into a narrated video — scene breakdown, image generation, voice and music — through a single multi-agent flow.

- Orchestrates GPT-4o for refinement and scene breakdown, DALL-E 3 for per-scene imagery, and ElevenLabs for narration.
- Composes the assets into a final video; containerized end-to-end with a TypeScript frontend and Express + MongoDB backend.

Stack: **Node.js, Express, React, Vite, TypeScript, OpenAI, DALL-E 3, ElevenLabs, MongoDB, Docker.**

Shorts-Gen

Solo builder · Video AI · 2025

Pipeline that turns long-form video into stylized 30-second shorts for TikTok / Reels / Shorts.

- ChatGPT identifies the most interesting segments; FFmpeg clips and re-encodes them; a style-transfer pass applies a chosen cartoon look.
- CLI-first, GPU-aware, optimized for vertical aspect ratios.

Stack: **Python, FFmpeg, OpenAI, Style transfer, CLI.**

Technical Skills

- **Data & MLOps:** Production Python (pandas, NumPy, Open3D, pytest); data-ingestion pipelines; queue-based orchestration (BullMQ; concepts transfer to Airflow / Prefect); vector stores (Qdrant, Pinecone); embeddings; Deepgram / Google STT; debugging + integrity tooling for ML teams.
- **Cloud & DevOps:** AWS (S3, IAM, Fargate), Google Cloud (Cloud Run, GCS), Azure Functions, Docker, Docker Compose, CI/CD (GitHub Actions, Jenkins, Railway), monitoring & logging.
- **Languages & Frameworks:** Python, TypeScript, Node.js, NestJS, Flask, FastAPI-style services, React, Next.js.
- **Storage & Search:** MongoDB, PostgreSQL, MySQL, CosmosDB, Redis, Elasticsearch, Kafka, Qdrant.
- **LLM tooling (working knowledge):** Multi-provider LLM gateways (OpenAI, Anthropic, Gemini) via LangChain; RAG; prompt fixers / evaluation loops; agentic pipelines.
- **Practices:** Internal tooling and dashboards for research teams, custom annotation platforms, end-to-end ownership, code reviews, mentoring.

Education

Bachelor's degree in Software Engineering

02/2012 - 06/2017

Code Academy – Mercy Corps program

2017

Languages

- Arabic (native)
- English (advanced)